Customer Churn Segmentation and Recommendation Memo

1. Project Overview

This project analyses customer information to identify customers who may require additional retention support. A rule-based risk scoring approach is used to prioritise customers based on tenure, support-ticket volume and monthly charges.

2. Methodology

The customer dataset was examined and relevant account and behavioural features were used to calculate a risk score. Customers were categorised into High, Medium and Low risk levels.
Customer segments were assigned using risk level, tenure and data-quality review requirements.
Records with missing critical information were flagged for review rather than being treated as fully reliable.

3. Customer Segmentation

The analysis groups customers into the following segments:

- **Data Review Required:** Customer records with missing critical information that needs to be checked.
- **New Customers Needing Support:** High-risk customers with 12 months or less of tenure.
- **High-Risk Customers:** Customers classified as high risk who do not fall into the new-customer segment.
- **Long-Tenure Customers:** Customers with 25 months or more of tenure who do not meet an earlier segment rule.
- **Regular Monitoring:** Remaining customers who do not meet the above conditions.

4. Key Findings

The notebook produces a customer-level risk table and a segment summary showing customer counts, average risk scores, tenure, monthly charges and support-ticket volume.
The actual segment counts and averages should be taken from the generated tables. The small sample size limits how confidently these findings can be generalised.
Historical churn rates should only be reported if a valid churn label is available in the dataset.

5. Business Recommendations

1. Review high-risk customers first and investigate their account or service issues.
2. Provide onboarding assistance and proactive follow-up for new customers identified as high risk.
3. Examine repeated support tickets to identify recurring service problems.
4. Verify missing or inconsistent data before making customer-level decisions.
5. Monitor customer segments over time and compare them with actual churn outcomes before relying on the risk scores for operational decisions.

6. Limitations

The approach uses manually defined rules and is not a trained predictive machine-learning model. The results depend on the selected thresholds and available data. A small dataset cannot establish that the scoring rules will generalise to a larger customer population.

7. Conclusion

The project demonstrates how customer data can be organised into interpretable risk segments to support retention planning. The resulting segment table and recommendations provide a starting point for further validation and customer-retention actions.